

Balraj Ashwath

Passionate about AI/ML Research |



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EDUCATION

- **National Institute of Technology Karnataka** Surathkal, IN
B.Tech in Electronics and Communications Engineering; GPA: 7/10 | Senior Thesis: 10/10 Aug. 2015 – Apr. 2019
- **Narayana Pre-University College** Bengaluru, IN
Karnataka State Pre-University; Percentage: 90% | JEE Mains: 176/360 (99th Percentile) Jul. 2013 – May. 2015
- **S Cadambi Vidya Kendra** Bengaluru, IN
Indian Certificate of Secondary Education (ICSE); Percentage: 92% | Java (CS): 100/100 Jul. 2010 – Apr. 2013

RESEARCH EXPERIENCE

- **Video Analytics Lab, CDS Department, Indian Institute of Science** Bengaluru, IN
Research Intern (Mentors: Prof. Dr. R. Venkatesh Babu & K. Ram Prabhakar) Jul. 2019 – Jan 2020
 - **High-Resolution HDR Deghosting using CNNs (Paper submitted to CVPR 2020):** Proposed a novel deep neural network to produce high-quality ghost-free HDR high-resolution images. The model is fast & achieves 6 seconds to fuse a sequence of three images with 16MP resolution. We achieve SOTA performance in terms of accuracy & speed on different publicly available datasets for High Dynamic Range Imagery.
 - **Self-supervised Learning Techniques for HDR Image Deghosting:** Ongoing Research.
- **Image Processing & Analysis Lab, National Institute of Technology Karnataka** Surathkal, IN
Undergraduate Researcher (Mentors: Prof. Dr. Shyam Lal & Karuna K. Eerapu) Jul. 2018 – Apr 2019
 - **Dense Refinement Residual Network for Road Extraction from Aerial Imagery Data:** Proposed a novel semantic segmentation architecture that effectively shares the learnt features across feature-spectrum & achieves a 2.74% increase in road accuracy on Massachusetts Roads dataset with ten-fold decrease in number of parameters.
 - **Dense Cascade Network for Road Extraction from Aerial Imagery:** Ongoing Research.

INDUSTRY EXPERIENCE

- **Microland Limited** Bengaluru, IN
Machine Learning Architect & Data Scientist Jul. 2019 - Present
 - **Service-Ticket Topic Modeling & Sentiment Analysis:** Unsupervised topic models (LDA/LSI) were developed to identify top topics in service tickets to automate tagging process. Implemented Sentiment Analysis models to improve ticket resolution process by augmenting feature set. [Tech Stack: Gensim; TF, PyTorch, ELK]
 - **Error Log Pattern Analysis, Clustering & Classification:** Hierarchical Clustering models were developed for unsupervised log pattern analysis using cosine similarity features, Ward's linkage criterion & Elbow methods. Supervised log classification models were implemented using LightGBM & CRFs on several standard log sources.
- **RBL Bank** Mumbai, IN
Computer Vision & Machine Learning Intern May 2018 – Jul. 2018
 - **e-KYC process Automation using CNNs for Optical Character Recognition on PAN/ Aadhar cards:** Developed CNN-based OCR models achieving 93% mean accuracy score & sub-40ms inference time per document.

PUBLICATIONS

- **Dense Refinement Residual Network for Road Extraction from Aerial Imagery Data**, Karuna K. Eerapu*, Balraj Ashwath*, Shyam Lal, Fabio D. Acqua & A.V. Narasimhadhan, *IEEE Access Journal*, 2019. [* - equal contribⁿ]

PROJECTS

- **Facial Keypoints Detection:** The objective is to predict landmark positions on face images. Applications include face recognition (biometrics), tracking faces, analysing facial expressions, etc. Accomplished best score among public kernels as measured by 2.03 RMSE score by using several data augmentation techniques to train a CNN model for regression.

- **Stack Overflow Tag/Score Prediction:** The objectives are to: 1) auto-tag questions by predicting their top tags 2) predict question scores based on text and tags. Developed multi-class classifiers using TF-IDF & XGBoost & compared with Random Forests, SGD & MultinomialNB classifiers. StackSample & StackLite datasets were used for the same.
- **Sentiment Analysis on Twitter & Amazon Reviews:** The objective is to predict sentiment of a tweet/review as either Positive, Neutral or Negative. Implemented model with Embedding + Bidirectional LSTM & achieved mean accuracy of 92.4%. Kaggle's Twitter Sentiment140 (1.6M) & Amazon Reviews (1M) datasets were used for the same.
- **Pneumothorax Segmentation:** The objective is to develop a model to classify & segment pneumothorax from a set of chest radiographic images. Implemented FC-DenseNet, (DeepLabV3+ & ResUNet) models & achieved dice-coefficient score of 0.84 on Kaggle's SIIM-ACR Pneumothorax Segmentation dataset.
- **AI Games:** Implemented AI games such as: 1) Flappy Bird using NEAT Framework 2) Tic-Tac-Toe using minimax algorithm. Used OpenAI Gym's RL environments such as MountainCar, Acrobot, etc to experiment with Q-Learning & Deep Q-Learning methods. Used Python's PyGame package to develop games like Tetris, Connect Four & Pong.
- **News Summarization with Attention:** The objective is to generate short length descriptions from news articles. Implemented sequence-to-sequence encoder-decoder LSTM models with attention on News Summary dataset.
- **Flickr Image Captioning:** The objective is to predict a sentence-based description for an image. Implemented models with pretrained ResNet50/VGG16 frontends as encoders & LSTM decoders using Flickr30K dataset for the same.
- **Toxic Comment Classification:** The objective is to identify & classify toxic online comments. Implemented model with GloVe Embeddings + BiLSTM achieving 97.8% AUC-ROC using dataset provided by Conversational AI Team.
- **DeepArt:** The objectives are to: 1) predict artist given an artwork 2) generate artwork using Neural Style Transfer. Classifier implemented using ResNet152 & Style Transfer Network using VGG19. [Dataset - Best Artwork of All Time]
- **Adversarial Domain Adaptation:** Implemented research papers in PyTorch. Models for papers such as Unsupervised Domain Adaptation by Backpropagation & Adversarial Discriminative Domain Adaptation were implemented.
- **Credit Card Fraud Detection:** The objective is to predict whether an online transaction is fraudulent. Performed EDA, feature aggregations & used LGBM model. Obtained dataset from IEEE-CIS Fraud Detection challenge.

PROGRAMMING SKILLS

ML Frameworks: TensorFlow, PyTorch, Keras | **Languages:** Python, Java, C, C++, JavaScript, R, Matlab, SQL, $\text{\LaTeX}$
Libraries/Tools: Pandas, OpenCV, Scikit-learn, SciPy, NumPy, SpaCy, NLTK, Gensim, Seaborn, BS4; Git, Photoshop

LICENSES & CERTIFICATIONS

Microsoft Certified: Azure AI Engineer Associate (AI-100); **Udacity:** Intel Edge AI (Nanodegree Program)
Coursera: Deep Learning Specialization (deeplearning.ai), IBM Data Science Professional Certificate.

RELEVANT COURSES

- **UG Coursework:** Data Structures & Algorithms, Pattern Recognition & Machine Learning, Statistical Analysis & Applications, Number Theory, Matrix Theory & Stochastic Processes, Cryptography & Data Security.
- **MOOCs:** Stanford's CS231n - Convolutional Neural Networks, CS234 - Reinforcement Learning, CS224N - Natural Language Processing; MIT's 6.S191 - Deep Learning, 6.S094: Deep Learning for Self-Driving Cars.

HONORS & AWARDS

Intel Edge AI Scholarship (2019), Merit-cum-Means Scholarship, FFE Scholarship, HackWithInfy Finalist (2018)

ACTIVITIES/INTERESTS & VOLUNTEER EXPERIENCE

- Reading Research Papers, Contributing on SO/Kaggle | Meditation, Mentoring Students, Marathon Running, Cooking
- Math/Science Tutor & Fundraising Volunteer at Prerana (NGO); Freelance Contributor at Google Crowdsourcing